

TidyTuesday Week 38

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Thursday 17th September, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to another thrilling installment of Randomly Yanking Two Time Series From FRED And Demanding They Explain Themselves. This week our tireless random-number generator paired up California house prices with discontinued financial investment flows, more California house prices with prime loan percentages, distillate fuel carbon emissions with Japanese retail stock sentiment, and U.S. assets with Asian business confidence. As always, none of these things have any earthly reason to be in the same room together, and yet here we are, running regressions and clutching our pearls at the results.

Monday kicked off with a bang: house prices in Oxnard-Thousand Oaks-Ventura versus some discontinued equipment-investment flow posted a raw R-squared of 0.87 and a p-value with seventy-five zeros of confidence behind it. Truly a relationship for the ages – until you detrend it, at which point the R-squared face-plants down to 0.23. In other words, two things that both went up over several decades were shocked, shocked, to discover they'd both gone up. It survives detrending in the technical sense that the p-value is still small, but it lost two-thirds of its explanatory swagger the moment we took away its shared upward escalator. Classic trend-mirage energy, only partially refunded. Tuesday's San Luis Obispo house prices versus prime loan sizes was refreshingly honest by comparison: it wasn't significant raw and it wasn't significant detrended. A rare series that admits it has nothing to say.

Wednesday is where things got genuinely funny. Distillate fuel commercial carbon emissions versus Japanese retail stock sentiment showed a raw positive relationship (coefficient 1.21, R-squared 0.48) that, upon detrending, flipped its sign entirely and became a negative relationship that's still significant. So depending on whether you remove the trend, burning distillate fuel is either good for Japanese retail confidence or bad for it. This is the statistical equivalent of a magic 8-ball that answers differently based on how hard you shake it, and I refuse to build a policy around it. Thursday closed us out with U.S. Assets versus Asian business confidence, which pulled the same disappearing-and-reappearing act: barely-there negative raw slope, then a significant positive slope once detrended. It technically holds up, but its coefficient is measured in something like 0.0000003, a number so small it needs a hug.

The usual reminder, delivered with all the gravity of a man reading a cereal box aloud: these are randomly paired economic series, thrown together by a script that does not know or care what any of them mean. Nothing here is causal, nothing here is rigorous, and any relationship that "survives" is surviving in a lab where spurious correlation is the house specialty. If two numbers drift upward for fifty years and we act surprised they correlate, the joke is entirely on us – which

is, of course, the whole point. See you next week for more numbers pretending to know each other.

2 Date: 2026-09-14

Series ID: ATNHPIUS37100Q

This series is titled All-Transactions House Price Index for Oxnard-Thousand Oaks-Ventura, CA (MSA) and has a frequency of Quarterly. The units are Index 1995:Q1=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1976-04-01 and the observation end date is 2026-04-01. The popularity of this series is 6.

Series ID: FBGFIEQ027S

This series is titled Financial business; gross fixed investment, nonresidential equipment, software, and structures, Flow (DISCONTINUED) and has a frequency of Quarterly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted Annual Rate. The observation start date is 1946-10-01 and the observation end date is 2017-10-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_FBGFIEQ027S	R-squared:	0.870
Model:	OLS	Adj. R-squared:	0.869
Method:	Least Squares	F-statistic:	1103.
Date:	Mon, 14 Sep 2026	Prob (F-statistic):	5.84e-75
Time:	17:07:08	Log-Likelihood:	-1926.0
No. Observations:	167	AIC:	3856.
Df Residuals:	165	BIC:	3862.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.341e+04	3837.676	3.493	0.001	5829.236	2.1e+04
value_fred_ATNHPIUS37100Q	745.7296	22.457	33.207	0.000	701.389	790.070

Omnibus:	1.131	Durbin-Watson:	0.039
Prob(Omnibus):	0.568	Jarque-Bera (JB):	0.803
Skew:	-0.148	Prob(JB):	0.669
Kurtosis:	3.165	Cond. No.	341.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

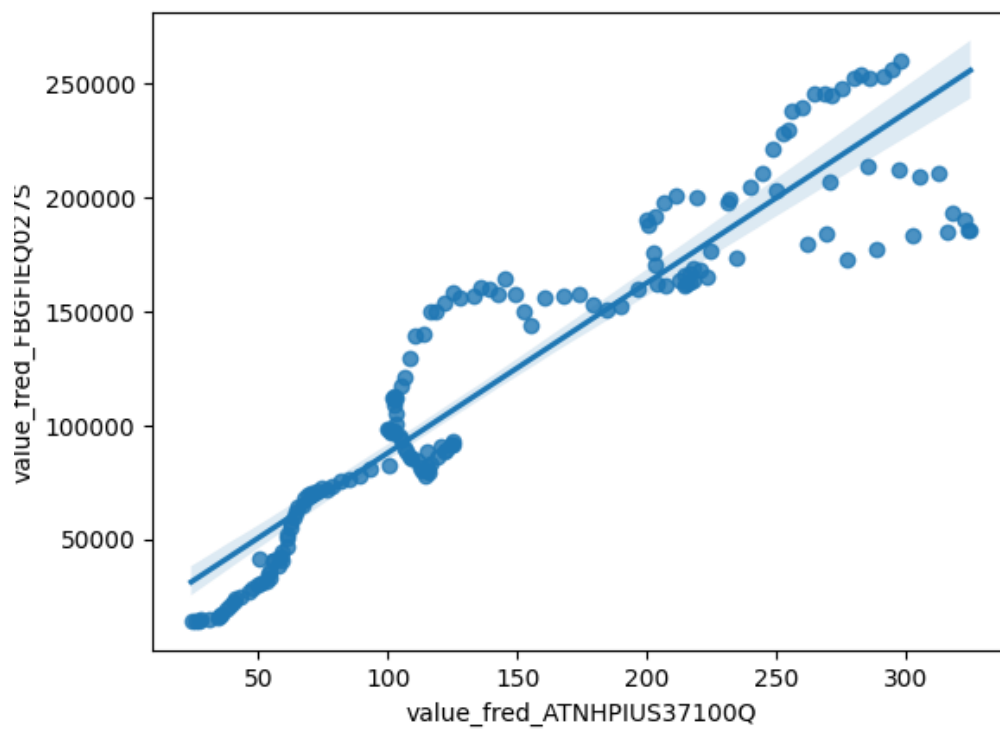


Figure 1: Raw Regression Plot for 2026-09-14

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_FBGFIEQ027S_detrended	R-squared:	0.233
Model:	OLS	Adj. R-squared:	0.228
Method:	Least Squares	F-statistic:	50.05
Date:	Mon, 14 Sep 2026	Prob (F-statistic):	4.06e-11
Time:	17:07:08	Log-Likelihood:	-1814.4
No. Observations:	167	AIC:	3633.
Df Residuals:	165	BIC:	3639.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-1.573e-09	985.135	-1.6e-12	1.000	-1945.095	1945.095
value_fred_ATNHPIUS37100Q_detrended	197.7479	27.952	7.075	0.000	142.559	252.937

Omnibus:	3.074	Durbin-Watson:	0.100
Prob(Omnibus):	0.215	Jarque-Bera (JB):	2.616
Skew:	-0.248	Prob(JB):	0.270
Kurtosis:	3.361	Cond. No.	35.2

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

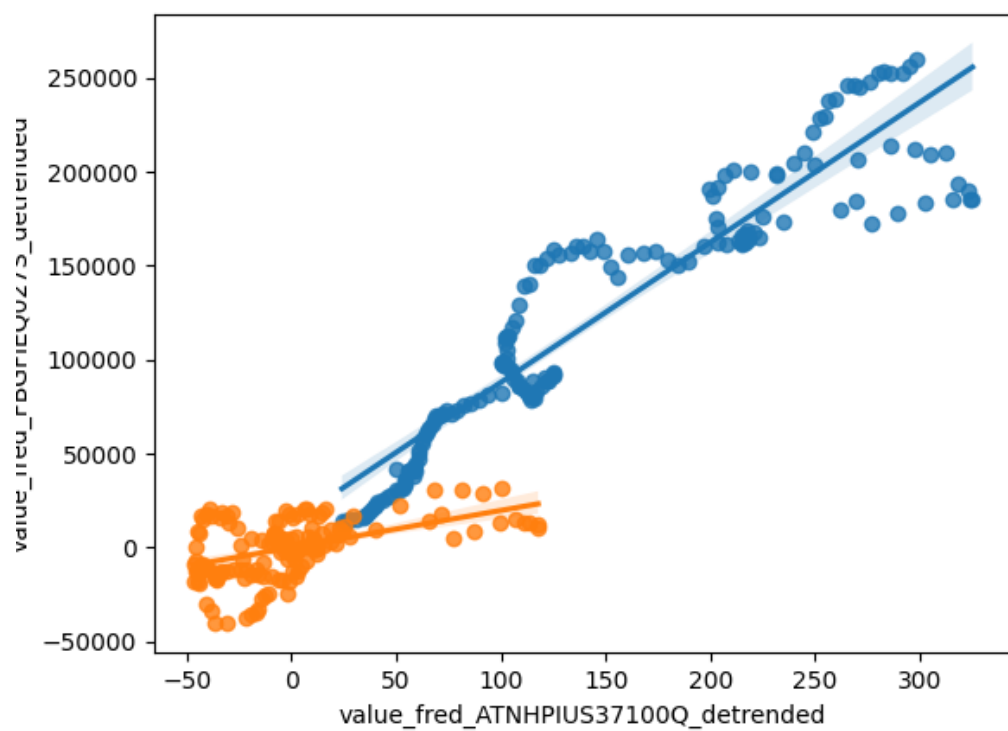


Figure 2: Detrended Regression Plot for 2026-09-14

3 Date: 2026-09-15

Series ID: ATNHPIUS42020Q

This series is titled All-Transactions House Price Index for San Luis Obispo-Paso Robles, CA (MSA) and has a frequency of Quarterly. The units are Index 1995:Q1=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1977-10-01 and the observation end date is 2026-04-01. The popularity of this series is 8.

Series ID: EARPRXDBNQ

This series is titled Percent of Value of Loans Prime Based by Average Loan Size (\$ thousands), Prime, Domestic Banks (DISCONTINUED) and has a frequency of Quarterly, 2nd Month's 1st Full Week. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1997-04-01 and the observation end date is 2017-04-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_EARPRXDBNQ	R-squared:	0.039
Model:	OLS	Adj. R-squared:	0.027
Method:	Least Squares	F-statistic:	3.241
Date:	Tue, 15 Sep 2026	Prob (F-statistic):	0.0756
Time:	15:46:24	Log-Likelihood:	-364.17
No. Observations:	81	AIC:	732.3
Df Residuals:	79	BIC:	737.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	176.5714	8.976	19.671	0.000	158.704	194.438
value_fred_ATNHPIUS42020Q	0.0686	0.038	1.800	0.076	-0.007	0.144

Omnibus:	0.239	Durbin-Watson:	1.271
Prob(Omnibus):	0.887	Jarque-Bera (JB):	0.081
Skew:	-0.077	Prob(JB):	0.960
Kurtosis:	3.017	Cond. No.	866.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_EARPRXDBNQ_detrended	R-squared:	0.027
Model:	OLS	Adj. R-squared:	0.015
Method:	Least Squares	F-statistic:	2.188
Date:	Tue, 15 Sep 2026	Prob (F-statistic):	0.143
Time:	15:46:25	Log-Likelihood:	-356.02
No. Observations:	81	AIC:	716.0
Df Residuals:	79	BIC:	720.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-4.581e-12	2.207	-2.08e-12	1.000	-4.393	4.393
value_fred_ATNHPIUS42020Q_detrended	-0.0708	0.048	-1.479	0.143	-0.166	0.024

Omnibus:	1.694	Durbin-Watson:	1.556
Prob(Omnibus):	0.429	Jarque-Bera (JB):	1.096
Skew:	0.241	Prob(JB):	0.578
Kurtosis:	3.303	Cond. No.	46.1

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

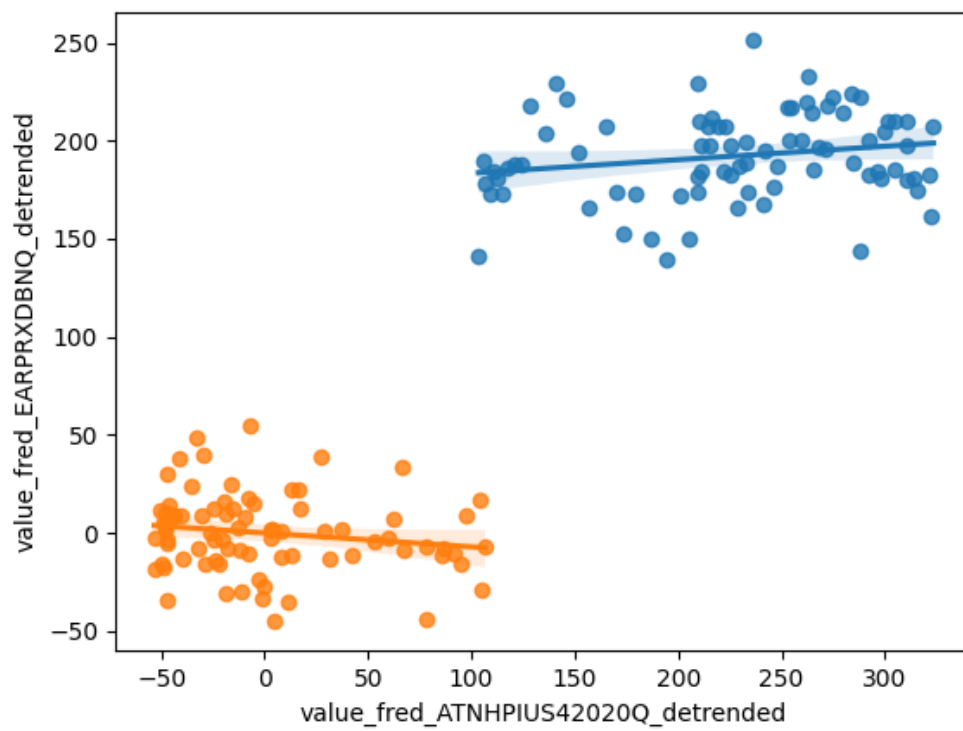


Figure 4: Detrended Regression Plot for 2026-09-15

4 Date: 2026-09-16

Series ID: EMISSCO2VDFCCBA

This series is titled Distillate Fuel Commercial Sector Carbon Dioxide Emissions and has a frequency of Annual. The units are Million Metric Tons CO2 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1973-01-01 and the observation end date is 2024-01-01. The popularity of this series is 1.

Series ID: BRVSLV02JPQ460S

This series is titled Business Tendency Surveys: Volume of Stocks: Economic Activity: Retail Trade, Except of Motor Vehicles and Motorcycles: Current for Japan and has a frequency of Quarterly. The units are Percentage balance and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1990-10-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_BRVSLV02JPQ460S	R-squared:	0.481
Model:	OLS	Adj. R-squared:	0.465
Method:	Least Squares	F-statistic:	29.65
Date:	Wed, 16 Sep 2026	Prob (F-statistic):	5.44e-06
Time:	15:39:06	Log-Likelihood:	-112.36
No. Observations:	34	AIC:	228.7
Df Residuals:	32	BIC:	231.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-20.4615	6.786	-3.015	0.005	-34.285	-6.638
value_fred_EMISSCO2VDFCCBA	1.2106	0.222	5.445	0.000	0.758	1.664

Omnibus:	6.618	Durbin-Watson:	1.027
Prob(Omnibus):	0.037	Jarque-Bera (JB):	5.041
Skew:	-0.838	Prob(JB):	0.0804
Kurtosis:	3.867	Cond. No.	178.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

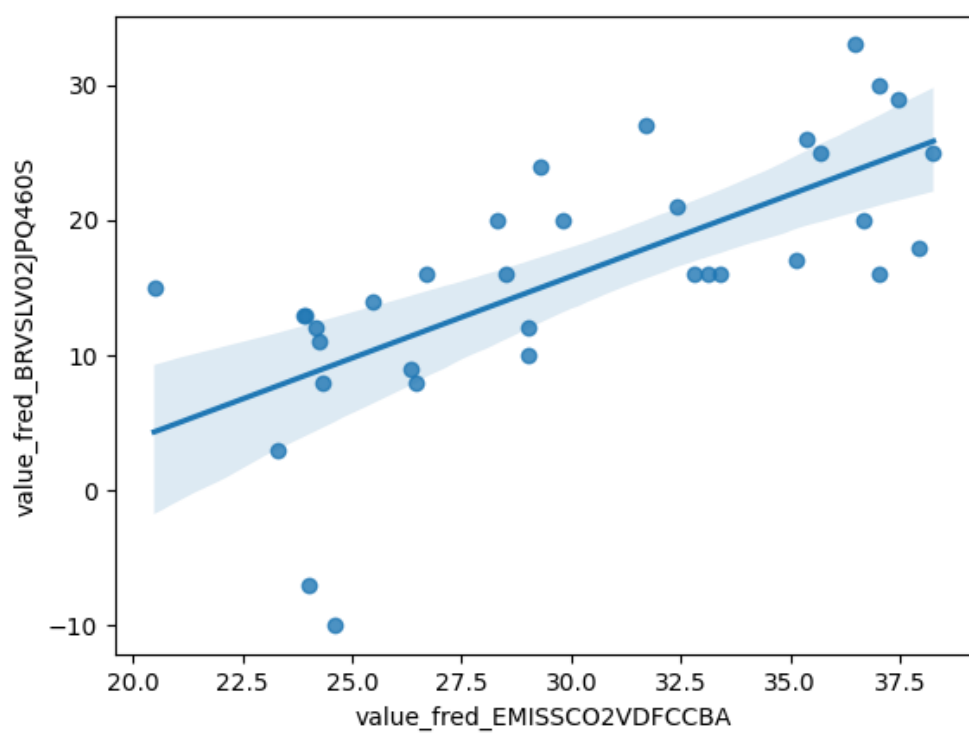


Figure 5: Raw Regression Plot for 2026-09-16

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_BRVSLV02JPQ460S_detrended	R-squared:	0.166
Model:	OLS	Adj. R-squared:	0.140
Method:	Least Squares	F-statistic:	6.379
Date:	Wed, 16 Sep 2026	Prob (F-statistic):	0.0167
Time:	15:39:06	Log-Likelihood:	-100.08
No. Observations:	34	AIC:	204.2
Df Residuals:	32	BIC:	207.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-6.513e-13	0.812	-8.02e-13	1.000	-1.654	1.654
value_fred_EMISSCO2VDFCCBA_detrended	-1.0600	0.420	-2.526	0.017	-1.915	-0.205

Omnibus:	7.112	Durbin-Watson:	1.687
Prob(Omnibus):	0.029	Jarque-Bera (JB):	5.544
Skew:	-0.868	Prob(JB):	0.0625
Kurtosis:	3.947	Cond. No.	1.93

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

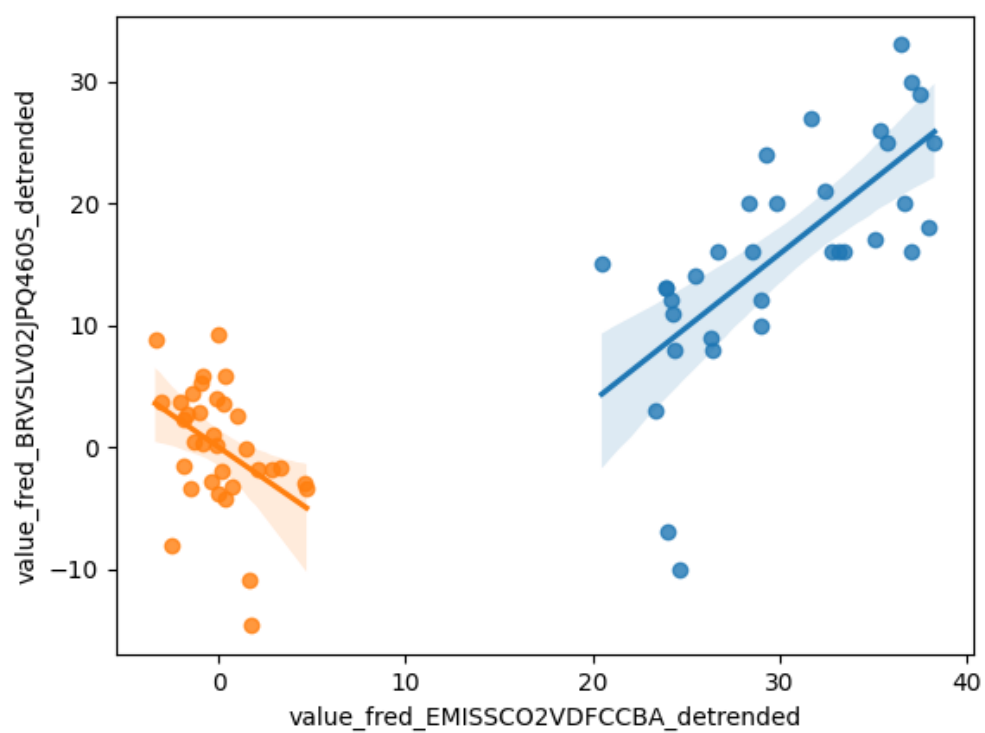


Figure 6: Detrended Regression Plot for 2026-09-16

5 Date: 2026-09-17

Series ID: IIPUSASSQ

This series is titled U.S. Assets and has a frequency of Quarterly, End of Period. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2006-01-01 and the observation end date is 2026-01-01. The popularity of this series is 10.

Series ID: BSCICP035AM665S

This series is titled Composite Leading Indicators: Composite Business Confidence Amplitude Adjusted for Major Five Asia Economies and has a frequency of Monthly. The units are Normalised (Normal=100) and the seasonal adjustment is Seasonally Adjusted. The observation start date is 2000-02-01 and the observation end date is 2023-12-01. The popularity of this series is 1.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_BSCICP035AM665S	R-squared:	0.083
Model:	OLS	Adj. R-squared:	0.070
Method:	Least Squares	F-statistic:	6.363
Date:	Thu, 17 Sep 2026	Prob (F-statistic):	0.0139
Time:	15:45:16	Log-Likelihood:	-123.79
No. Observations:	72	AIC:	251.6
Df Residuals:	70	BIC:	256.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	101.6727	0.802	126.848	0.000	100.074	103.271
value_fred_IIPUSASSQ	-8.045e-08	3.19e-08	-2.522	0.014	-1.44e-07	-1.68e-08

Omnibus:	13.288	Durbin-Watson:	0.449
Prob(Omnibus):	0.001	Jarque-Bera (JB):	16.146
Skew:	-0.826	Prob(JB):	0.000312
Kurtosis:	4.628	Cond. No.	1.25e+08

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.25e+08. This might indicate that there are strong multicollinearity or other numerical problems.

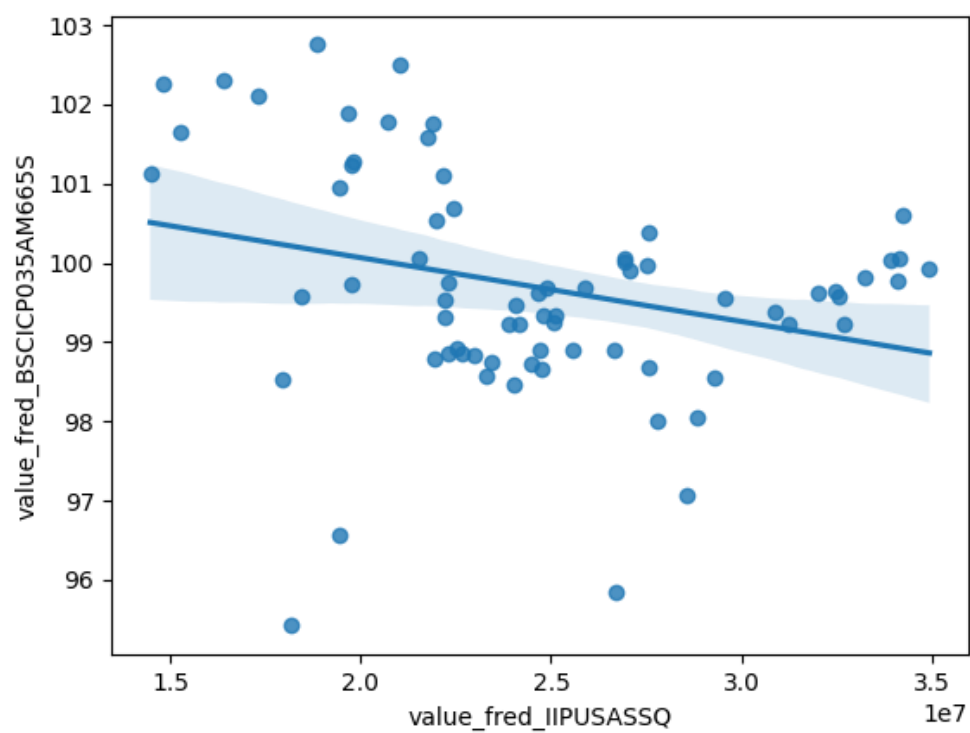


Figure 7: Raw Regression Plot for 2026-09-17

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_BSCICP035AM665S_detrended	R-squared:	0.178
Model:	OLS	Adj. R-squared:	0.167
Method:	Least Squares	F-statistic:	15.21
Date:	Thu, 17 Sep 2026	Prob (F-statistic):	0.000219
Time:	15:45:16	Log-Likelihood:	-112.66
No. Observations:	72	AIC:	229.3
Df Residuals:	70	BIC:	233.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-2.082e-12	0.138	-1.51e-11	1.000	-0.276	0.276
value_fred_IIPUSASSQ_detrended	3.494e-07	8.96e-08	3.900	0.000	1.71e-07	5.28e-07

Omnibus:	26.580	Durbin-Watson:	0.538
Prob(Omnibus):	0.000	Jarque-Bera (JB):	53.882
Skew:	-1.260	Prob(JB):	1.99e-12
Kurtosis:	6.407	Cond. No.	1.54e+06

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.54e+06. This might indicate that there are strong multicollinearity or other numerical problems.

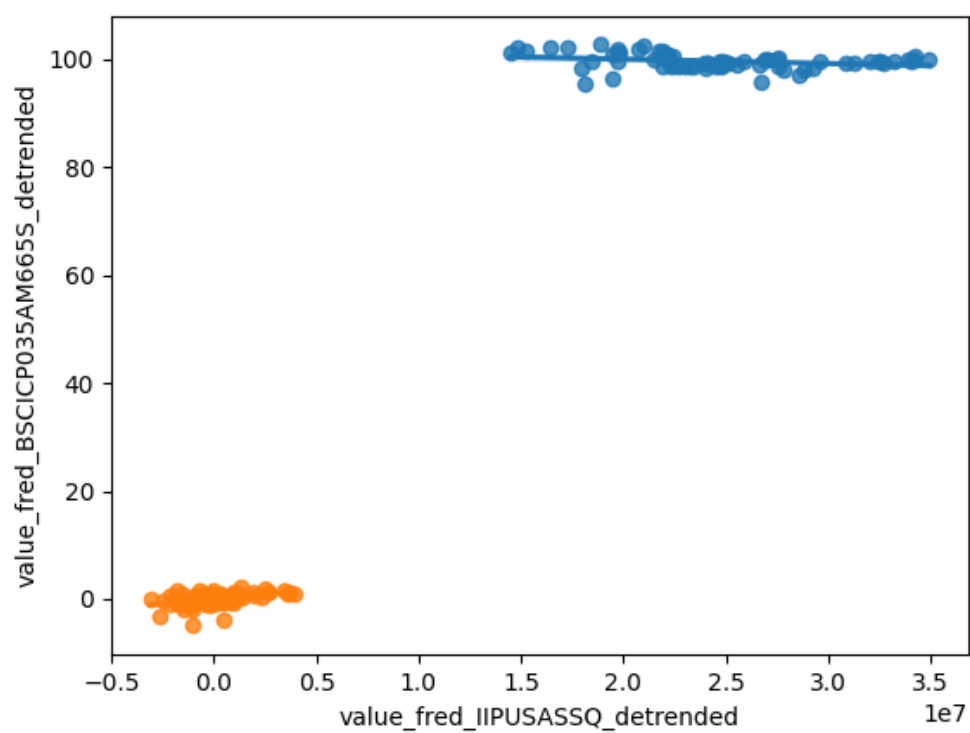


Figure 8: Detrended Regression Plot for 2026-09-17